

Smart City Bike Sharing Analysis – One-Page Summary Report

Project Overview

This project analyzes Smart City Bike Sharing data using Power BI to evaluate station performance, bike availability, utilization efficiency, and operational status across multiple European cities. The dataset was cleaned and transformed using Power Query, modeled using a star schema, and visualized through an interactive dashboard to support data-driven decision-making.

Key Findings

- The bike-sharing network consists of approximately **1,000 stations**, **57,000 bike stands**, **20,000 available bikes**, and **941 active stations**.
- **92.79%** of stations are operational (OPEN), while **7.21%** are currently CLOSED, indicating high overall service availability.
- **Bruxelles-Capitale** and **Lyon** have the highest number of available bikes, suggesting greater bike availability in these cities.
- **Jcdcauxbike** records the highest utilization rate (**66%**), followed by **Lund** and **Ljubljana**, indicating efficient bike usage.
- The bike-sharing system spans **20+ European cities**, demonstrating broad geographic coverage and supporting sustainable urban mobility.

Dashboard Highlights

The interactive Power BI dashboard includes:

- KPI cards for Total Stations, Total Bikes, Total Bike Stands, Utilization Percentage, and Active Stations.
- Top 10 Cities by Available Bikes.
- Top 10 Cities by Utilization Rate.
- Station Operational Status (Open vs. Closed).
- Geographic map showing bike station locations.
- Station-level detail table with availability and occupancy metrics.
- Interactive filters for city, year, quarter, month, and station status.

Conclusion

The Smart City Bike Sharing dashboard provides a comprehensive overview of network performance and station operations. It enables stakeholders to monitor key performance indicators, identify high- and low-performing cities, and make informed decisions to improve service quality, operational efficiency, and the overall user experience.